Lahore Grammar School
Johar Town Senior Boys Campus

# CYGNUS' VEIL

Study Guide




## 11 . 12 . 13 SEPTEMBER

LAHORE GRAMMAR SCHOOL JOHAR TOWN FOR BOYS
INNOVENTIONS '26
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**Round 1:**
**CYGNUS' FEUD**
Duration: 2 hours
Maximum Team Members: 3
Minimum Team Members: 2

Teams will face off against each other one at a time. All rules of conventional Family Feud apply here (participants will be briefed once before the round).
The teams will be provided with buzzers; the first team to press the buzzer gets to answer the question.
In between the questions, there will be interruptions in the form of surprise questions.

The contestants will have to guess the top 10 most popular objects/ideas from the following topics:

- Astronomy in fiction
- Nebulae
- Deep Sky Objects (DSOs)
- Constellations
- Astrophysics ideas/theorems

**Non-elimination, all points are added up to round 2.**

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**Round 2:**
**THE ENDURANCE**
Duration: 3 hours
Maximum Team Members: 3
Minimum Team Members: 2

This round will consist of two parts:

**Part I:**
Delegates will be given 10 exoplanets. However, only one ship will be sent from Earth, with the most suitable candidate from these exoplanets to be studied.

Teams will receive information in the form of observations made through different instruments, each instrument providing different kinds of data, for example:
- stellar observatory observations providing stellar activity.

Teams will be given the option to choose from multiple data values or observations specific to their exoplanets. Teams are advised to choose wisely as they will not be given any data other than what they have opted for.
Delegates will not only be considering habitability in their assessment of the exoplanets, rather their choices should be based on the balance between the 3 following factors:

1. Habitability
2. Accessibility (Communications and resource availability)
3. Colonization potential

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The teams will also be assigned hidden objectives, which will be strictly specific to the team itself. For example, Mission A, assigned to Team #01 in our database, will be to maximize the ability to conduct research on specific biological phenomena on their exoplanet.

The idea is that our spacecraft only has enough fuel to visit one planet; therefore, delegates will have to nominate one planet as the most suitable after consideration of each aforementioned factor.

Please note: There will be no correct or incorrect choice. Choices will be assessed on the basis of reasoning and appropriate consideration of each team's requirements

**Part II:**
The delegates will then participate in an auction, where they get the opportunity to purchase materials that help them with their missions in some way, shape, or form, being given 2,000,000 credits in order to do so. In order to ensure an intense environment between teams, materials will have overlaps between missions (they'll be important to more than one mission).

The teams will then have to make a 2-minute video that explains how their materials work in tandem with their plan to accomplish their mission on their respective planet.

**This round is an elimination round.**
**Score weightage is as follows:**

- **Round 1: 30%**
- **Round 2- Part I: 40%**
- **Round 2- Part II: 30%**

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**Round 3:**
**OPERATION HELLSTORM**
Duration: 2 hours
Maximum Team Members: 2
Minimum Team Members: 2

A large celestial map is displayed, containing targets such as planets, moons, stars, nebulae, black holes, exoplanets, and asteroids. Teams choose a target they want to capture and are given an astronomy challenge based on that object.

A correct answer allows them to claim the target and earn its points. However, teams can also challenge targets already controlled by another team.

Upon success in solving the challenge, the team will be allowed steal that target and its points. This can be done if the other team(s) do not answer correctly.

As the round progresses, higher-value and more difficult targets are unlocked, creating opportunities for teams that are behind to catch up. Certain special targets will also be included; specifics will be shared prior to the round.

The round continues until the map is exhausted and/or the time runs out, with the live celestial map and leaderboard constantly running.

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**The team with the highest score at the end wins Operation Hellstorm and Cygnus Veil.**
**The teams would be marked on their accuracy when answering the questions, their speed and their calculations.**

**Topics to prepare for Cygnus' Veil:**

1. Basic astrophysics
2. Astronomy in pop culture
3. Modern space engineering and its basic principles
4. Astronomical data processing and its inferences
5. General astronomy facts
6. Observational Astronomy

**Use of AI is strictly prohibited throughout the course of the category.**

**Please note:**
**The category heads have the authority to change how the category and its rounds are conducted whenever they deem it necessary or appropriate.**